Biochemistry Cume
September 3, 2004

1. (15pts each) The alpha-galactosidase exhibits maximum enzyme activity at pH 4.5 and is stable over a broad range of pH. Its pI is 6. You decided to batch purify the enzyme using the DEAE (diethylaminoethyl) Sephadex which has a working pH range of approximately 2 to 9. The crude protein buffered at pH 8 was successfully batch purified. What would be the effect on the batch purification of alpha-galactosidase if

a) the ionic capacity of the gel was increased?
b) the pore size of the gel was decreased?
c) the gel was changed to a strong anion exchanger?
d) the gel was changed to a beaded cation exchange gel such as carboxymethyl Sephadex?

2. (20 pts each) Let's assume that you isolated alpha-galactosidase from newly isolated fungi using a combination of anion exchange and gel filtration chromatographies. You learned that the molecular weight of the protein was 600,000 daltons using the gel-filtration.

a) Since you were fully aware that yeast alpha-galactosidase was a glycoprotein and that the newly isolated protein might also be a glycoprotein, you removed the carbohydrates with glycosidases and ran another gel-filtration to learn that it has the molecular weight of 550,000 daltons. How many amino acid residues are there in the protein? (Average amino acid's weight is 110 daltons).
b) You suspected that the enzyme might be a multimer. You ran the SDS-PAGE of the unglycosylated protein in the absence of beta-mercaptoethanol and learned that it migrates as 92,000 daltons. In the presence of beta-mercaptoethanol it migrates as two peaks, 50,000 and 42,000 daltons. Explain the results concisely.